

PHE-31 Internal Hydrotest and Tightening Limits Reference

Document code: PHE31-LIM-2026-05. This reference defines internal task limits and record-handling rules for the PHE-31 pre-job workpack data set.

Limit table

limit_id	asset_tag	measurement_type	limit_type	value_low	value_high	unit	handling_note
LIM-001	PHE-31	tightening_dimension_A	range	424.0	428.0	mm	compare same-unit compression measurements only
LIM-002	PHE-31	tie_bolt_torque	range	180.0	195.0	Nm	M24 lubricated tie bolts
LIM-003	PHE-31	nozzle_flange_bolt_torque	range	80.0	90.0	Nm	M16 flange bolts
LIM-004	PHE-31	hydrotest_pressure	exact	9.5	9.5	bar	do not compare with operating pressure
LIM-005	PHE-31	hydrotest_duration	minimum	30.0		min	hold duration minimum
LIM-006	PHE-31	pressure_drop	maximum		0.2	bar	allowed drop during hydrotest hold
LIM-007	PHE-31	pressure_gauge_range	exact	0-16	0-16	bar	use explicit gauge range text
LIM-008	PHE-31	gauge_calibration_age	maximum		30	days	age measured from certificate date to test date
LIM-009	PHE-31	plate_count	exact	160	160	plates	current rebuilt plate pack count
LIM-010	PHE-31	gasket_material	exact	NBR	NBR	text	current gasket material

Record-handling rules

rule_id	rule_text
R-001	Blank numeric, permit, calibration, closeout, or source-reference fields must be treated as MISSING.
R-002	Descriptive notes such as weep, stain, odor, or condition observations are TEXT_ONLY unless a number and unit are recorded.
R-003	Do not convert units unless a source file explicitly provides the converted value.
R-004	Hydrotest pressure in bar must not be compared with gauge range text.
R-005	Pressure drop during hydrotest must be compared against LIM-006 only.
R-006	Tightening dimension A must not be compared with plate count.
R-007	For duplicate rows, comparison_source_file must equal source_file.

PHE-31 hold point reference

hold_point_id	asset_tag	checkpoint	required_record	source_required	note
HP-001	PHE-31	pre-isolation	LOTO reference	yes	LOTO reference must not be blank
HP-002	PHE-31	pre-disassembly	blind list complete	yes	all blind references required
HP-003	PHE-31	plate pack rebuild	tightening dimension A	yes	record in millimetres
HP-004	PHE-31	hydrotest setup	gauge calibration date	yes	calibration must be explicit
HP-005	PHE-31	hydrotest hold	pressure drop	yes	record bar drop after hold time
HP-006	PHE-31	final inspection	leak check observation	yes	text-only if not quantified